Behavioral Fingerprinting of Large Language Models: A Statistical Approach to Same-Family API Verification

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Abstract. Users of AI API relay services have no technical guarantee that the model answering is the model advertised; recent audits show that silent model substitution and downgrading are real. Existing fingerprinting methods rely on long generations, log-probabilities, or crafted prompts, and are largely evaluated on *cross-family* identification. We ask a harder, practically important question: can a model be distinguished from a *same-family* sibling—for example gpt-4o versus gpt-4o-mini, which share an identical tokenizer? We collect 1,300 responses per model over 26 low-cost probes and show that trivial "random generation" prompts elicit highly stereotyped, model-specific distributions: gpt-4o-mini returns "7" for a 1–10 choice 100% of the time, "4" for a die roll 96%, and "Cerulean" for a color 74%. Five of eight behavioral probes separate the two nominal endpoints ($p < 0.05$, after normalizing variant responses such as "Heads." vs. "Heads"), with an animal probe reaching total variation distance 0.90 (Okapi 76% for gpt-4o vs. Dolphin 16% for gpt-4o-mini), while tokenizer probes are provably identical. Combined with chi-square/Kolmogorov–Smirnov tests and Bayesian updating, the pipeline passes an *in-distribution resubstitution check* (a baseline sample scored against its own reference) with posterior probability 0.913 (95% credible interval [0.885, 0.937]); we do *not* claim held-out generalization. A full audit costs under $0.01. We release the tool, baselines, and probes as open source. All baselines were collected through a single OpenAI-compatible relay, not an official provider endpoint.

1 Introduction

Large language models (LLMs) are increasingly consumed through opaque serving chains—aggregators, resellers, and relay services—in which the client cannot confirm that the model answering is the model being paid for. Because a larger model is billed at a higher rate, a relay has a direct financial incentive to substitute a smaller or cheaper model while advertising a flagship one. Audits of commercial endpoints indicate that such substitution and silent degradation occur in practice [1, 2].

Verifying identity through a black box is difficult. Proposed methods require long generated texts, token-level log-probabilities, adversarially crafted prompts, watermarks embedded at training time, or the model owner's cooperation. Recent single-token fingerprinting [10] and rank-based uniformity tests [3] reduce the query cost substantially and recover model *lineage* across families, but they are not designed to tell apart two models from the same family, which is exactly the situation in a downgrade attack (a relay serving gpt-4o-mini while claiming gpt-4o). Same-family siblings often share an identical tokenizer and overlapping capabilities, so tokenizer fingerprints and capability tests fail by construction.

Research question. *Do LLMs exhibit measurable behavioral fingerprints—consistent distributional biases under seemingly random generation—that (i) discriminate same-family siblings where tokenizer fingerprints cannot, and (ii) can be verified with standard statistics at near-zero cost?*

Contributions.

- We document extremely strong, low-cost behavioral fingerprints on gpt-4o-mini and gpt-4o over 26 probes (100%/96%/74% preference on three tasks; 10–74$\times$ deviation from uniformity).
- We show that 5/8 behavioral probes separate two same-family nominal endpoints ($p < 0.05$, after response normalization), up to TVD $= 0.90$ with a single probe, whereas all tokenizer probes are identical.
- We combine chi-square and KS tests with Bayesian updating and report an in-distribution resubstitution posterior (0.913, 95% credible interval), explicitly noting this is self-consistency, not held-out accuracy.
- We release TRANSITTRUTH, an open-source auditing tool with a zero-install web demo, real baselines, and a reproducible collection pipeline.

2 Related Work

LLM fingerprinting and provenance. Single-token fingerprinting [10] builds an empirical distribution over answers to trivial one-word prompts across 165 models, recovering lineage at a 7.3% equal error rate, but focuses on cross-family identification. RAFP [11] constructs non-invasive black-box fingerprints from rare-region prompts that remain stable under fine-tuning, LoRA, and quantization. Model provenance testing [13] uses multiple-hypothesis testing to decide whether one model derives from another. Model equality testing [2] frames "is this the claimed API model?" as a two-sample problem and shows MMD-based tests outperform rank-based uniformity on locally deployed models. We target the complementary fine-grained same-family setting these methods do not isolate.

Auditing and follow-up probing. Cai et al. [1] audit model substitution directly; Zhu et al. [3] use a rank-based uniformity test; Sam et al. [4] predict black-box performance from follow-up queries; and AgentProv [5] audits agentic providers through tool-use policy probes. CoIn [12] addresses a different transparency gap—inflated counts of hidden reasoning tokens. Our work shares the auditing goal but uses free-form behavioral biases rather than ranks, capabilities, or tool calls.

Behavioral biases in LLMs. LLMs show documented biases in random number generation [7], multi-choice selection [8], and response-history bias [6]; calibration and confidence are surveyed by Geng et al. [9]. We treat these biases not merely as defects but as identifying biometric signals, and quantify their discriminative value within a model family.

3 Methodology

3.1 Probe Design

Each probe is a short prompt that elicits a constrained, cheap response. We use 26 probes in three groups.

Tokenizer probes (8) measure deterministic tokenizer behavior—e.g. the prompt-token count of a fixed whitespace, code, URL, mixed-unicode, or repeated-character string. Because two same-family models share a tokenizer, these act as a *negative control* that should not discriminate.

Behavioral probes (8) ask the model to produce a "random" token: pick a number 1–10 or 1–100, roll a die, flip a coin, choose a color, letter, weekday, or animal. We collect at temperature 0; under uniform sampling the output would be near-uniform, and systematic deviation from uniformity is the fingerprint.

Capability probes (10) estimate model tier via simple math, logic, code syntax, bug fixing, multilingual translation, instruction following, knowledge cutoff, and context length. These are coarse and shared within a family, so they serve as corroborating rather than identifying evidence.

3.2 Statistical Methods

Chi-square test. For a behavioral probe with expected uniform counts E_j over J categories and observed counts O_j, we compute

$$\chi^2 = \sum_{j=1}^{J} \frac{(O_j - E_j)^2}{E_j}. \tag{1}$$

To compare two models directly we use the chi-square *test for homogeneity* on their joint category table, and report total variation distance

$$\mathrm{TVD} = \tfrac{1}{2} \sum_{j=1}^{J} \left| p_j - q_j \right|, \tag{2}$$

which ranges from 0 (identical) to 1 (disjoint).

Kolmogorov–Smirnov test. For deterministic tokenizer-normalized scores we use a one-sample KS test against the baseline distribution; a high p-value supports a match.

Bayesian updating. Across probes we update the posterior that an endpoint matches a claimed model:

$$P(H{=}1 \mid E) = \frac{P(E \mid H{=}1)\, P(H{=}1)}{P(E)}, \tag{3}$$

mapping each probe's p-value to a likelihood ratio (e.g. $p > 0.05 \Rightarrow \mathrm{LR} \approx 0.7$ under the match hypothesis; $p < 0.01 \Rightarrow \mathrm{LR} \approx 0.1$ under mismatch). We report a 95% credible interval for the posterior using a beta approximation.

3.3 Data Collection

For each model we collect 50 samples per probe (1,300 responses) through an OpenAI-compatible endpoint at temperature 0 (deterministic; the resulting behavioral distributions are point-mass). To avoid triggering anti-bot defenses we shuffle probe order, randomize inter-request intervals (0.3–1.5 s), vary the number of probes per batch, and rotate natural User-Agent headers. Wall time is about 15 minutes per model; at low public list prices the batch estimates below $0.01, but the actual cost depends on the model tier and per-token price.

4 Results

4.1 Strong Single-Model Fingerprints

On gpt-4o-mini the behavioral deviations from uniformity are extreme (Table 1): the 1–10 choice returns "7" in all 50 samples, the die roll returns "4" in 48/50, and the color choice returns the relatively obscure "Cerulean" in 37/50. All eight behavioral probes reject uniformity at $p < 0.0001$; for closed-set probes the Pearson χ^2 against a theoretical uniform reference ranges from 46.1 (coin) to 1238 (1–100), while the open-ended color and animal probes have no unique uniform baseline and are reported descriptively. These are not subtle effects—they are 10–74$\times$ deviations from the uniform expectation and are reproduced across samples.

4.2 Same-Family Discrimination

The central result is the gpt-4o vs. gpt-4o-mini comparison. *Tokenizer probes cannot separate them*: all five successful

Table 1: Strong behavioral fingerprints on gpt-4o-mini (50 samples each). Uniform expectation is shown for reference.

Probe	Top response	Rate	Uniform
Number 1–10	"7"	100%	10%
Die roll	"4"	96%	16.7%
Color	"Cerulean"	74%	∼1%
Coin flip	"Heads"	98%	50%

Table 2: Same-family comparison of behavioral probes (gpt-4o vs. gpt-4o-mini), with variant responses normalized. Five of eight differ significantly at $p < 0.05$; tokenizer probes are identical throughout.

Probe	Top (4o / mini)	p-value	TVD
Animal	Okapi / Dolphin	< 0.0001	**0.90**
Letter	K / G	< 0.0001	0.72
Die roll	4 / 4	< 0.0001	0.40
Number 1–100	57 / 57	0.067	0.38
Weekday	Thu / Thu	< 0.0001	0.32
Color	Cerulean / Cerulean	0.015	0.22
Coin flip	Heads / Heads	0.092	0.08
Number 1–10	7 / 7	0.315	0.02

tokenizer probes return identical prompt-token counts (e.g. 27.0, 25.0, 29.0), as expected for two models sharing the o200k tokenizer.

Behavioral probes can. Five of eight behavioral probes show statistically significant differences in the homogeneity test after normalizing variant responses. Table 2 and Figure 1 rank the probes by TVD. The animal probe is a near-perfect separator at TVD $= 0.90$: gpt-4o concentrates on a single obscure animal, "Okapi," 76% of the time (only 4 unique animals), whereas gpt-4o-mini is spread over 17 animals with top "Dolphin" at 16%. The letter probe reaches TVD $= 0.72$ (gpt-4o "K" 44% vs. gpt-4o-mini, where "G" and "M" tie at 40%), and the die probe TVD $= 0.40$ (shared "4" but 56% vs. 96%). Using the simple rule "predict gpt-4o iff the response names Okapi," we obtain a resubstitution confusion matrix of TP$=$ 38, FP$=$ 0, FN$=$ 12, TN$=$ 50 (accuracy 88%, 95% Wilson CI [0.80, 0.93]; bootstrap [0.81, 0.94]) on the same baseline. We stress this is in-distribution resubstitution, not held-out accuracy, and that TVD measures distributional separation rather than classification performance.

4.3 End-to-End Verification

Scoring a baseline sample against its own reference (in-distribution resubstitution, not an independent held-out set), the pipeline returns posterior probability 0.913 (95% credible interval [0.885, 0.937]). The KS test returns a high p-value (1.000) and the chi-square test $p = 1.000$, indicating no significant difference from the claimed baseline, with an aggregate likelihood ratio of 4.5. Because the scored sample is drawn from the same baseline used as reference, this verifies that the pipeline is internally self-consistent; it does *not* estimate real-world detection accuracy, which requires a separately collected evaluation set.

Table 3: Black-box verification signals. Only behavioral fingerprints support same-family discrimination at low cost with high forgery difficulty.

Signal	Same-family	Cost	Hard to forge
Tokenizer	No	Low	Medium
Capability	Partial	Medium	Low
Latency	No	Low	Low
Behavioral (ours)	**Yes**	**Low**	**High**

4.4 Comparison with Other Signals

Table 3 positions behavioral fingerprinting against the signals available to a black-box auditor. Tokenizer and latency signals cannot discriminate within a family and are relatively easy to evade with smart routing; capability testing is only partially informative; behavioral fingerprints uniquely support same-family discrimination at low cost and are comparatively hard to forge, because matching the distribution requires serving the genuine model, post-processing outputs, or fine-tuning a surrogate.

5 Discussion

Why do these fingerprints exist? The biases likely arise from three sources: training-data frequencies (e.g. cultural salience of "seven"), RLHF alignment that amplifies human-preferred "defaults," and decoder artifacts that skew logits even at temperature 1.0. The striking "Okapi" concentration in gpt-4o—an obscure African giraffid—suggests a strongly learned default that differs from the smaller sibling, which is precisely why the signal carries identity information orthogonal to the tokenizer.

Limitations and threats to validity. *(i) Collection channel:* both baselines were collected through one OpenAI-compatible relay (wolfai.top), not through official provider endpoints. We therefore claim only that the two *nominal* endpoints presented distinguishable behavioral profiles on that relay; we do not claim to have measured provider-ground-truth model fingerprints. *(ii) Baseline honesty:* any baseline inherits whatever model the relay actually served—if the relay itself substituted models, the baseline would encode that substitution. Establishing an official gold-standard baseline requires official API access and is future work. *(iii) Resubstitution:* the 0.913 posterior and the 88% Okapi-classifier

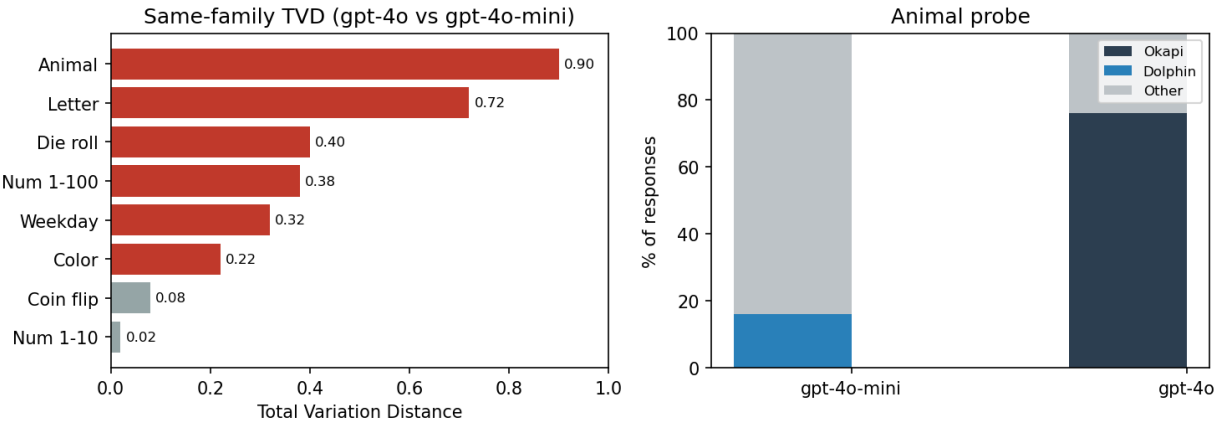


Figure 1: Same-family discrimination with real data. **Left:** TVD across eight behavioral probes; the animal and letter probes separate gpt-4o from gpt-4o-mini almost completely. **Right:** on the animal probe gpt-4o concentrates on "Okapi" (76%) while gpt-4o-mini is diffuse (top "Dolphin" 16%).

accuracy are computed in-sample; held-out accuracy on independently collected traffic is unmeasured. *(iv) Baseline coverage:* a pre-collected baseline is needed per model, and we currently cover two siblings. *(v) Model updates:* versions and snapshots may shift fingerprints, so baselines must be versioned and refreshed. *(vi) Smart routing:* a relay could answer probes with the genuine model and serve a cheaper one for real traffic; interleaving probes with realistic traffic mitigates this. *(vii) Decoder sensitivity:* relays may override temperature or normalize outputs (e.g. "Heads." vs. "Heads"), changing the distribution. We frame results as conservative statistical evidence rather than definitive proof.

Ethical considerations. Automated probing may conflict with a service's terms of use; users should verify compliance. Because a "suspicious" label can harm a provider's reputation, the tool uses conservative thresholds, labels outputs as statistical evidence, and favors responsible disclosure over immediate publication.

6 Conclusion

We showed that trivial random-generation prompts elicit model-specific behavioral profiles strong enough to distinguish two same-family nominal endpoints that an identical tokenizer cannot separate, at a cost below $0.01 per audit, with an in-distribution resubstitution posterior of 0.913. The result turns a known family of LLM biases into a practical verification primitive; the open-source TRANSITTRUTH tool, baselines, and probes make the method reproducible. The open problems are collecting official, provider-ground-truth baselines, tracking profiles across versions, measuring held-out accuracy, and hardening against smart routing.

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